Ibukun Adeniji

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As a **data analyst** with an expertise in both **machine learning and deep learning**, and hands-on experience in **embedded systems**, I am eager to secure a fulltime position to apply my skills in a real-world environment. My experience includes working as an algorithm engineer in the perception team for the **Aggies Autonomous Auto**, sponsored by **General Motors and SAE International**, which has equipped me with valuable industry-specific knowledge. I also got the chance to work as a data science at **Electric Power Research Institute (EPRI)** where I improved my abilities in model development and data-driven decision-making while learning more about real-world uses of data science in the energy sector. I aim to bring my technical proficiency and data-driven insights to a forward-thinking organization, addressing real-world challenges and making a significant impact.

EDUCATION AND HONORS

***North Carolina A & T State University*, Greensboro, North Carolina** — Dec. 2024

Master of Science, Electrical and Computer Engineering — GPA: 3.72

Relevant Courses: Machine Learning and Data Mining CSE 805, Signal Processing ECEN 641, Embedded Systems Design ECEN 621

***Federal University Oye-Ekiti*, Ekiti State, Nigeria**. — Dec. 2018

Bachelor of Engineering (Honors), Electrical and Electronics — GPA: 3.44

TECHNICAL SKILLS

Tools and Frameworks: Python (PyTorch, TensorFlow, OpenCV, Keras, NLTK, Scikit-learn, NumPy, Matplotlib, Seaborn, Pandas, Databases (MySQL), Version control (Git/GitLab), Recommendation engine, Predictive modeling, Pretrained models, Generative AI, and Large Language Models.

Data Visualization/Statistical Software: Tableau, PowerPoint, Power BI, SPSS, Excel, MATLAB

Platforms: PyCharm, Jupyter Notebook, Visual Studio Code

Soft Skills: Communication, Presentation, Problem Solving and Project Management

WORK EXPERINCE

Data Science Intern | Electric Power Research Institute (EPRI), Knoxville, TN, USA — Jun. 2024 – Aug. 2024

- Developed a transformer-based autoencoder model for unsupervised clustering of power quality events, enabling precise feature extraction and robust clustering of complex voltage waveform data, significantly enhancing the accuracy of event classification in electrical grids.
- Optimized clustering performance by applying K-Means on the latent space and using PCA for dimensionality reduction, achieving a silhouette score of 0.905. Presented well defined clusters using t-SNE visuals; with attention scores revealing the model's emphasis on critical event regions.
- Delivered insights through attention heatmaps, improving event detection and authored technical documentation that enhanced knowledge transfer.

Data Analyst | Ibadan Electricity Distribution Company, Ibadan, Nigeria — May. 2019 – Jun. 2020

- Applied Python and ETL techniques to extract and clean data from different sources, including SQL servers and flat files, achieving a 55% reduction in data cleaning time.
- Collaborated on predictive models (XGBoost, Decision Trees, and LR) to forecast energy consumption patterns, reducing revenue losses by 33%.
- Utilized data visualization tools: Power BI and Excel to produce visually appealing dashboards and reports for stakeholders to enhance decision
- Authored and maintained comprehensive documentation on database schema, energy consumption models, and SQL scripts on GitHub, ensuring team-wide clarity and seamless future reference.

Graduate Assistant | Dr Sunil's lab | North Carolina A&T State University, Greensboro, NC, USA — Aug. 2023 – Present

- Led the implementation of AI-driven models for automatic nuclei segmentation on rat brain images using convolutional neural networks (CNNs) with TensorFlow, Keras, and OpenCV, achieving a SSIM score of 93% compared to ground truth IR images.
- Developed custom data augmentation pipelines to enhance model robustness and generalization across varied imaging conditions.

PERSONAL PROJECTS

Animal Sounds Classification System using Convolutional Neural Networks (Kaggle)

- Developed and implemented CNN models for the classification of animal sounds, achieving high accuracy in distinguishing between species-specific vocalizations. Achieved Impressive results with an accuracy of 0.95, recall of 0.94, and precision of 0.9.
- Preprocessed audio data by extracting relevant features using Mel-Frequency Cepstral Coefficient, deployed on Streamlit cloud for user access.

Predicting Customer Churn in Telecom with Decision Tree

- Built a Python-based Machine Learning model using Decision Tree to predict customer churn for a telecom company, achieving 93% accuracy.
- Conducted comprehensive exploratory data analysis to identify churn drivers and utilized the SMOTE algorithm to tackle data imbalance.

CERTIFICATION

- Machine Learning with Python (Cousera) — May. 2021
- Natural Language Processing with Deep Learning in Python (Udemy) — Dec. 2021
- Data Analytics Consulting Virtual Internship (KPMG) — Feb. 2023
- Lean Six Sigma Green Belt — April. 2024